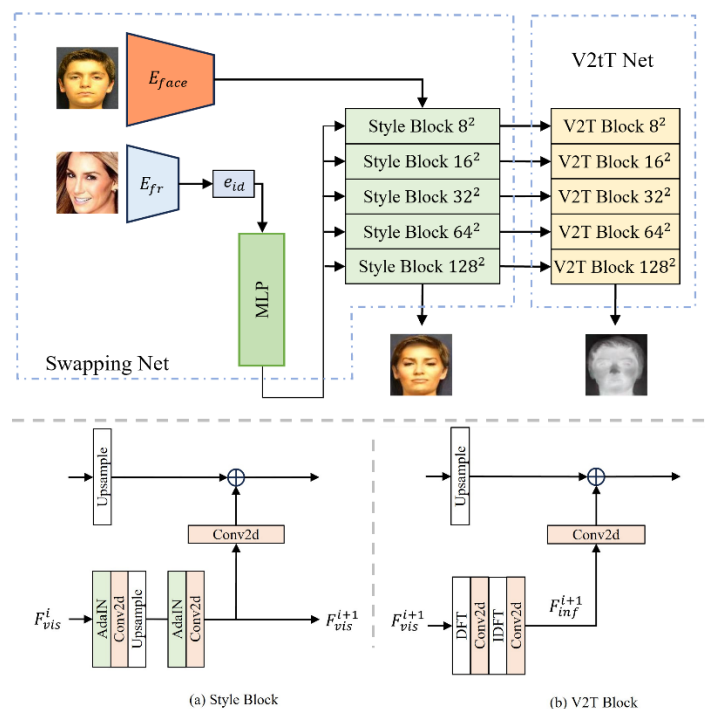


Dual-stage generation for heterogeneous face recognition

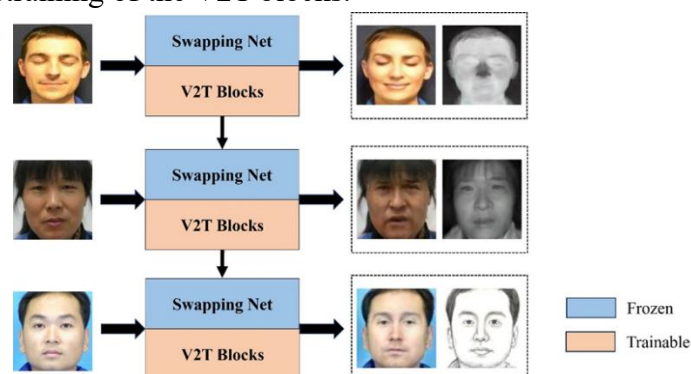
This repo is a PyTorch implementation of **Dual-stage generation for heterogeneous face recognition**. The Dual-stage generation method is used to generate paired heterogeneous faces for heterogeneous face recognition. The framework of this method is as follows:



The Swapping Net is designed to realize the face swapping function. It extracts identity information from the CelebA dataset and replaces the identity information of visible faces in the Tufts Face Dataset. The V2tT Net translates visible-domain features to thermal-domain features and generates thermal images that share the same identity information as the swapped visible images. In this manner, the identity information of the CelebA dataset is introduced into the Tufts Face Dataset.

An interesting experiment

Benefits from the two stages strategy, this method is extensible to other facial modality with only the retraining of the V2T blocks.



Some results



Source Images (Top Row); Images Generated by Our Method (Bottom Row)

Metrics:

Average FID: 28.756

The performance of training LightCNN as DVG-Face method:

Rank-1: 60.98%, VR@FAR=1%:44.51%, VR@FAR=0.1%:16.27%

The performance of training LightCNN by generating images with specific class labels (the code will be released at DSG-HFR):

Rank-1: 83.41%, VR@FAR=1%:62.84%, VR@FAR=0.1%:37.09%

Requirements

Download pretrained model SimSwap from <https://github.com/neuralchen/SimSwap>, then unzip and put it to the folder `./pretrained_model/simswap`.

Download ArcFace model from the above website. Then put it to the folder `./pretrained_model`.

Download pretrained model Semi-HFR from <https://github.com/xiyin11/Semi-HFR> and put it to the folder `./pretrained_model`.

Download Tufts face dataset and unzip it. Then, put the visible face images to the folder `./tufts_dataset/vis` and thermal face images to `./tufts_dataset/inf`.

Download CelebA dataset and unzip it. Then, put all images to the folder `./celebA_dataset/data`.

Training process

Before training or generation, please verify the correctness of all file paths referenced within the program.

1. Train Swapping Net

Run `train_face_swapping.py`

2. Train V2t Net

Run `train_v2tt.py`

Generate images

Run `generation.py`. The results will be saved in the folder `./generated_images`.